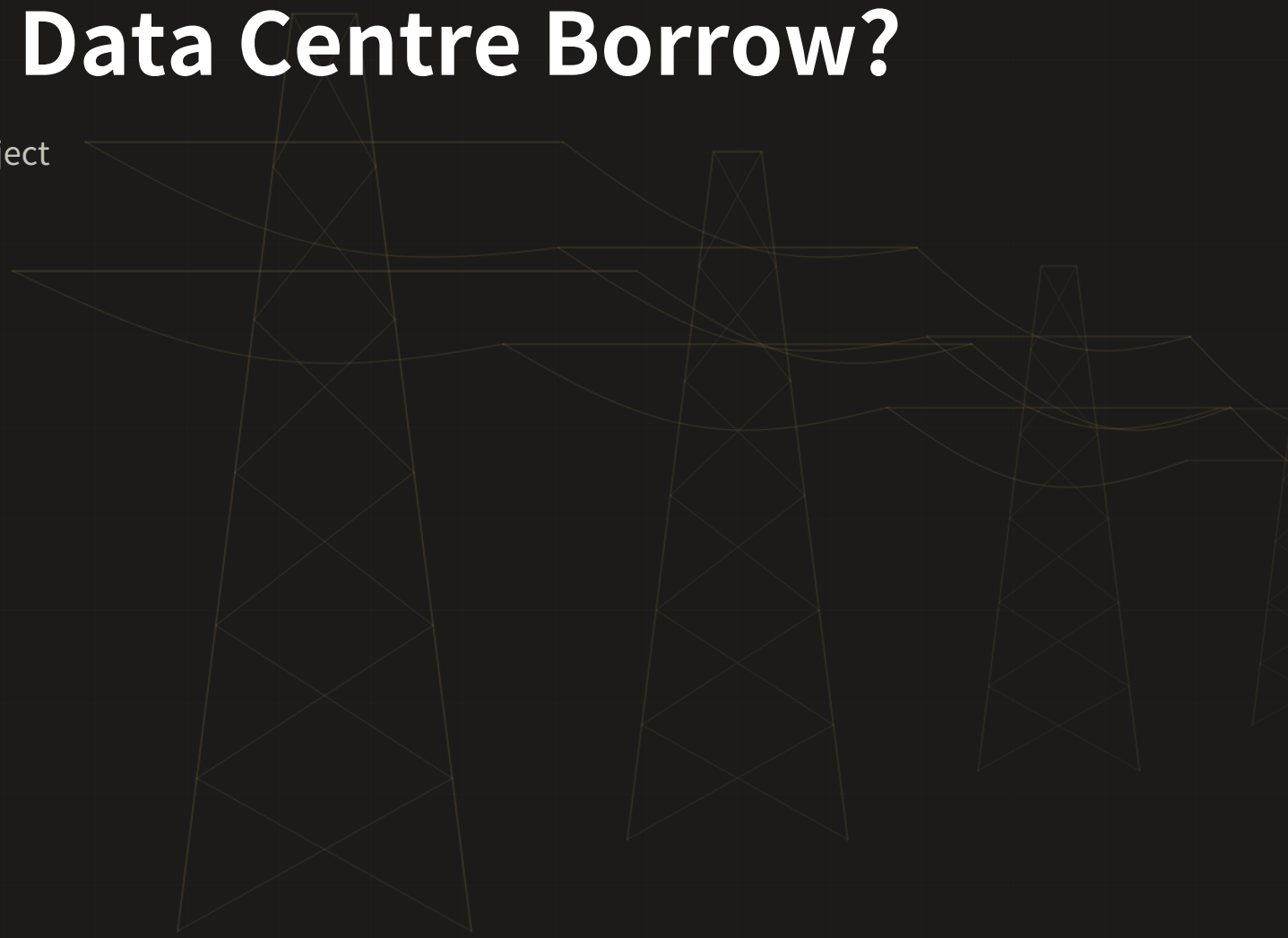
KSJ RESEARCH

Research Note Vol.2

How Much Can an AI Data Centre Borrow?

A lender's view of a 40 MW Korean and a 100 MW U.S. project



Shinjae Kang

2026.10

KSJ Research · For education and research only · Not investment advice · Not a real project

Korea: grid (3.0% approved) and pre-leasing (16–23.5%); U.S.: scale and rates (UST 10y 0.91pp above KTB 10y) — the first drivers of debt capacity

One PF engine (sculpting, waterfall, DSRA, LLCR, IRR) run on a 40 MW Korean (KRW) and a 100 MW U.S. (USD) case

Key metrics [*1,*2,*3,*4,*5,*6]

Key metric	Value	p.
Seoul-metro grid approval (2024.8~2026.3)	**3.0%**	1
Seoul-metro pre-leasing (2027–28 deliveries)	16–23.5%	1
Demand growth IEA 2024→2035 (low · base · high)	4.9% · 10.1% · 13.7%	2
KTB 3y · 10y (2026-10-20)	3.84% · 4.18%	2
UST 10y (2026-10-16)	**5.09%**	2
Peer-implied equipment price rise (low · mid · high)	3.3% · 10.9% · 23.9%	2
Korea 40 MW debt capacity	after inputs	3
U.S. 100 MW debt capacity	after inputs	6
Phase 2 (20 MW) option value	after inputs	5

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified) *2 Press citing market reports (rent, pre-leasing; source being verified) *3 IEA, Energy and AI (Apr 2025), executive summary *4 Bank of Korea ECOS market rates (daily), USD/KRW *5 FRED, Federal Reserve Bank of St. Louis (DGS10) *6 FSS DART annual reports (consolidated)
Note: KSJ Research. Hypothetical cases; definitions and sources in footnotes and A2–A3. Judgements are the author's own

3.0%

Seoul-metro phase-1 grid approval (2024.8~2026.3)

Korea 40 MW — grid approval and pre-leasing bind first

- 1,010 MW approved of 33,592 MW applied in the Seoul metro, phase 1 (3.0%)
- Pre-leasing 16–23.5% for 2027–28 deliveries; rent about KRW 250,000/kW-month
- Filling at market growth (IEA base 10.1%/yr) takes 16.8 yrs to full

5.09%

UST 10y (2026-10-16, FRED)

U.S. 100 MW — demand is large; base rates are higher

- Base-rate gap: UST 10y 5.09% − KTB 10y 4.18% = +0.91pp
- Virginia commercial sales 5.6%/yr vs U.S. 0.9% (2015–2025, EIA)
- U.S. commercial average price 13.41¢/kWh (2025); power is passed through to the tenant

+10.9%

Three peers' combined implied price rise (FY2022A→FY2025A)

Cost risk — peer margins 5.2%→14.4%, evidence of pricing power

- If all margin gain were price: implied rise 3.3%–23.9% (upper bound, not causal)
- CAPEX impact = power-equipment share × implied price rise (computed once the share is sourced)

Contents

Structure → data → sizing → stress → option → comparison → implications: each section's conclusion

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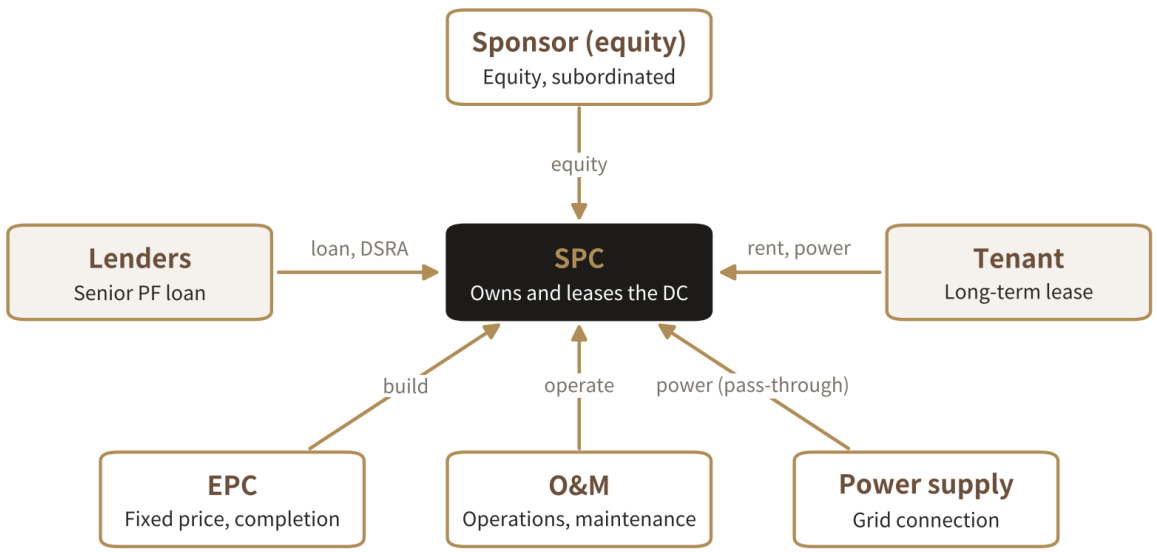
VII **Implications** 7
Implication: for Korean AI data-centre PF, 'when' comes before 'how much' — grid approval and an anchor tenant first

App. **Appendix** A1–A6
A1 Q&A · A2–A3 assumptions and sources by case · A4 Data notes · A5 Methodology · A6 Glossary

Structure: an SPC with long-term committed leases and power pass-through — Korea 20 MW + 20 MW option, U.S. 100 MW

Case A (main) 40 MW outer Seoul metro, KRW; Case B (comparison) 100 MW Virginia, USD — 1,010 MW approved of 33,592 MW applied in the Seoul metro, phase 1 (3.0%)

The SPC repays debt from long-term committed rent; power is passed through to the tenant *1



Source: *1 KSJ Research
Note: Case A structure (hypothetical). Power pass-through 100%: SPC's net power cost = 0

Five contracts shape the cash flow — lenders watch leasing, rates, cost and grid *1

Contract	Party	Key terms	Lender risk
Lease	Tenant	Long-term committed lease; rent per kW-month; annual escalation	Pre-lease ratio; tenant credit
Loan	Lenders	Target DSCR; max gearing; tenor, grace; DSRA	CFADS volatility; rates
EPC	Contractor	Fixed price; completion date; delay LDs	Equipment prices; delay
Power supply	Utility	Grid-connection approval; capacity	Phase 2 grid approval
Power cost	Tenant	Usage × tariff, 100% pass-through	Pass-through carve-outs

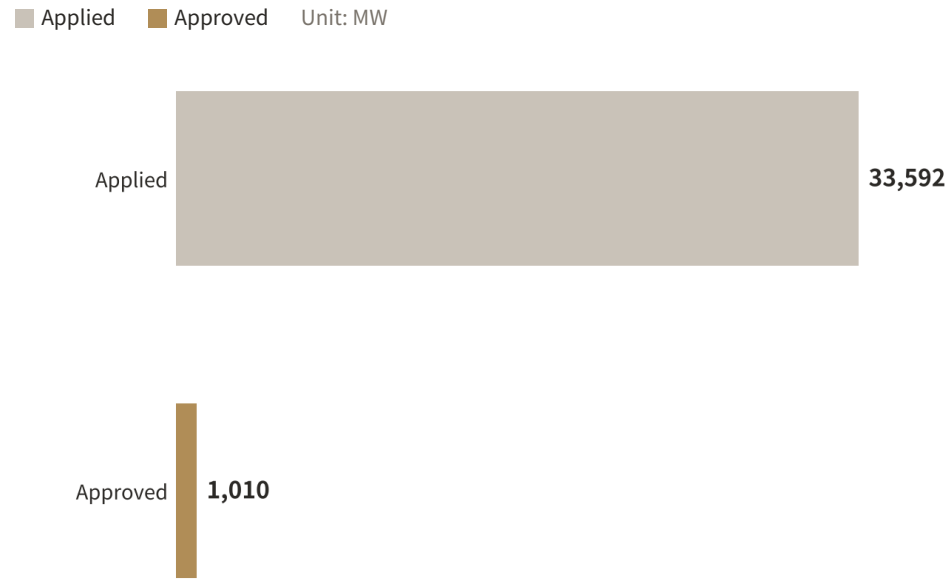
Source: *1 KSJ Research
Note: Hypothetical structure; actual terms vary by project

Korea builds in 20 MW steps, the U.S. in one 100 MW block — same engine, different parameters *1,*2,*3,*4,*5

	Case A (main)	Case B (comparison)
Location	Outer Seoul metro (Incheon, W. Gyeonggi)	Virginia, U.S.
Capacity (IT load)	40 MW = 20 MW firm + 20 MW option	100 MW, single phase
Currency · base rate	KRW · KTB 3y/10y (3.84%/4.18%)	USD · DGS10 (5.09%)
Rationale	Grid approval 3.0%; grid impact assessment from 10 MW; domestic operating scale ~40 MW	Hyperscale scale; Virginia commercial sales growth (EIA)

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified) *2 Distributed Energy Act (grid impact assessment) *3 Bank of Korea ECOS market rates (daily), USD/KRW *4 FRED, Federal Reserve Bank of St. Louis (DGS10) *5 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Not real projects. Base rates: latest business day on or before 2026-10-20

Only 3.0% of phase-1 grid applications approved — why phase 2 is an option *1

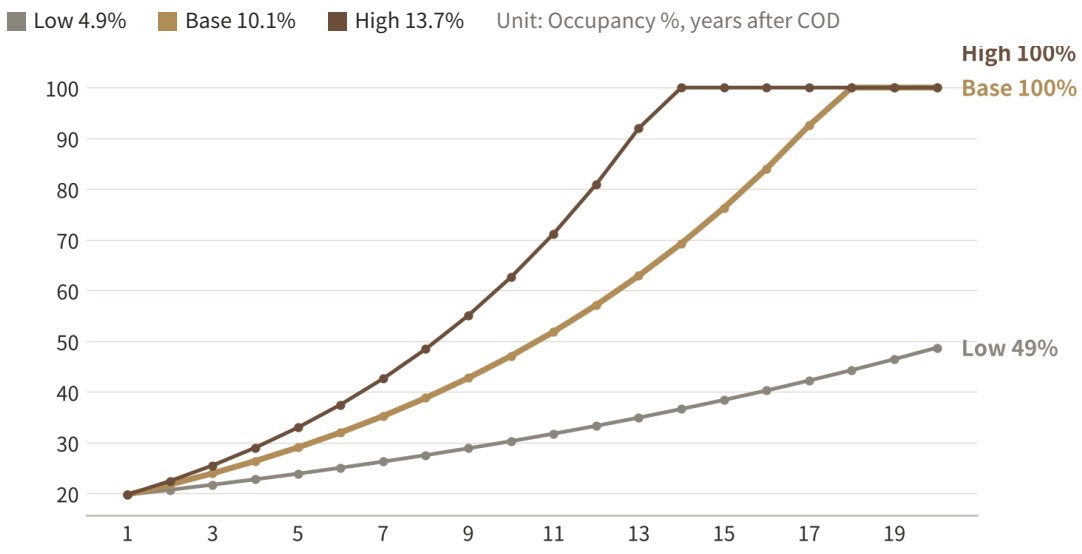


Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified)
Note: 2024.8~2026.3. Ratio = approved ÷ applied (KSJ Research). Projects ≥10 MW need a grid impact assessment

Vol.1 data into PF inputs: base demand growth 10.1%, UST 0.91pp above KTB, peer margins point to pricing power

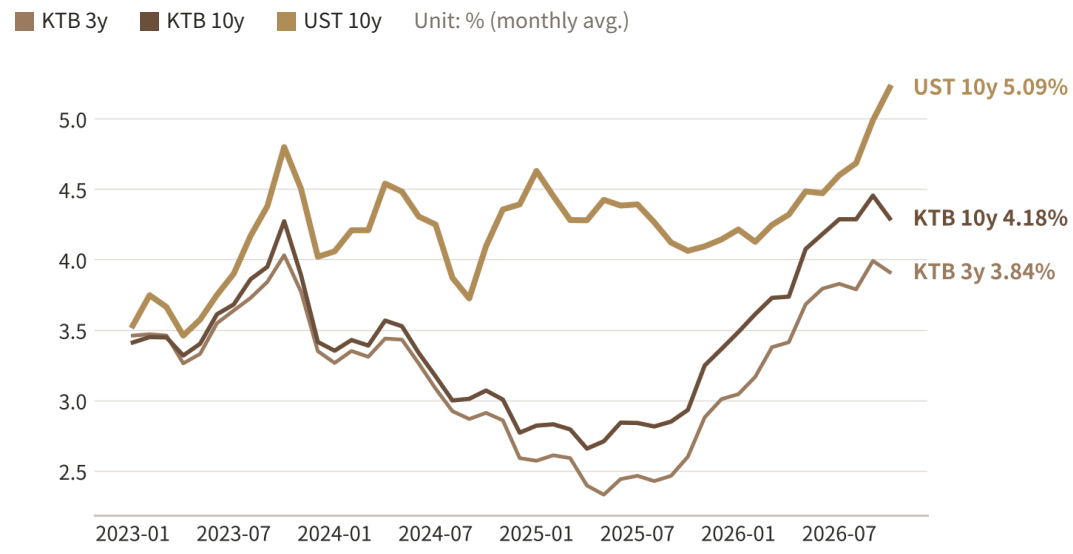
Demand (IEA) → ramp-up · rates (ECOS, FRED) → base rates · peer margins (DART) → CAPEX stress · Virginia (EIA) → Case B demand and power

Filling at market growth, pre-lease 19.8% reaches full in 16.8 yrs (base) — lenders count contracted rent *1,*2



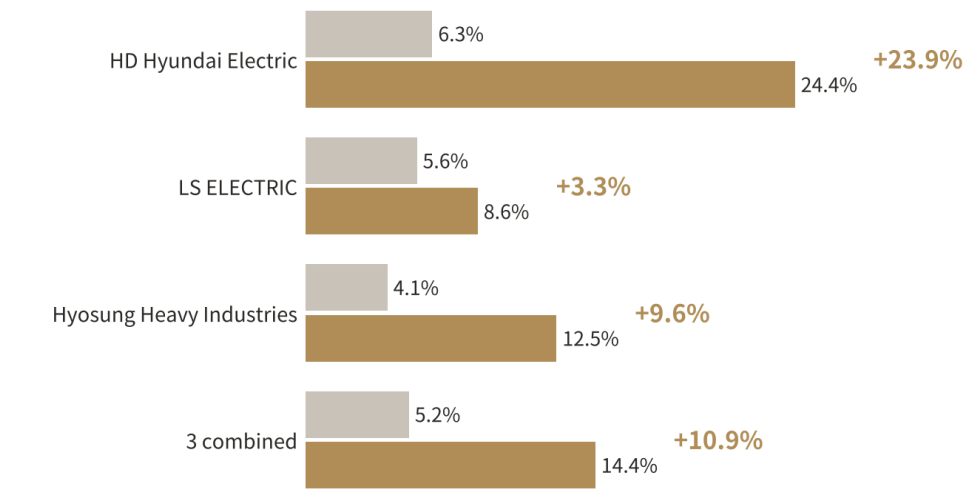
Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Press citing market reports (rent, pre-leasing; source being verified)
Note: Growth = IEA data-centre power CAGR 2024→2035 (415→low 700, base 1,200, high 1,700 TWh). Occupancy = pre-lease × (1+growth)^t; capped at 100% until stabilised occupancy is sourced

Base rates (2026-10-20): KTB 3y 3.84%, 10y 4.18%; UST 10y 5.09% — 0.91pp higher *1,*2



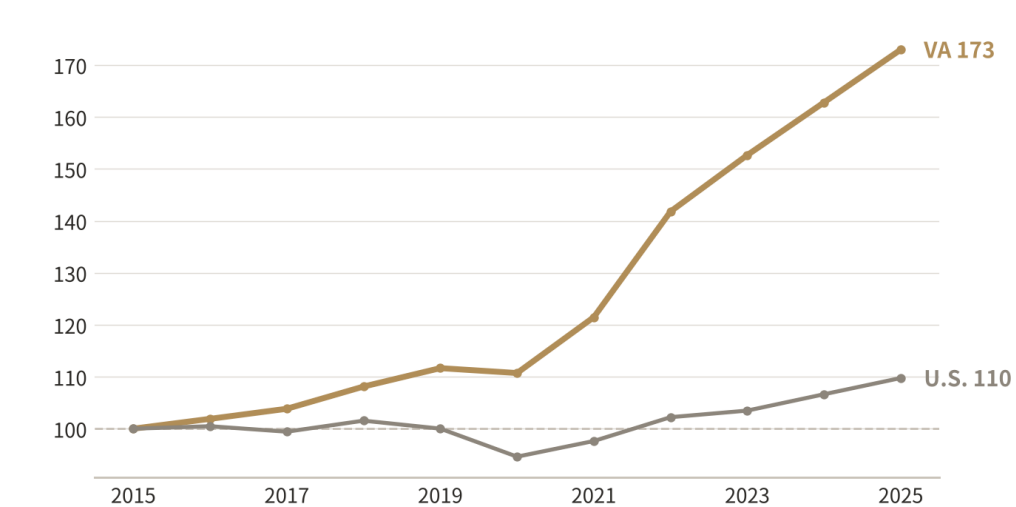
Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Lines = monthly averages; end labels = as-of values. Model base rates: Korea = KTB 3y (construction), 10y (term); U.S. = DGS10 (2026-10-16). Margins are separate

Peer margins 5.2%→14.4% — if all price, equipment +10.9% (3.3–23.9%) *1



Source: *1 FSS DART annual reports (consolidated)
Note: Implied rise = (1 − start OPM) ÷ (1 − end OPM) − 1: an upper bound if unit costs were flat and all margin gain came from price. Volume, FX and mix are not separated — evidence for a stress, not causation

Virginia commercial sales 5.6%/yr (U.S. 0.9%) — U.S. commercial price 13.41¢/kWh (2025) *1



Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Demand evidence for Case B (Virginia 2025 +6.2% y/y). Power-cost sensitivity uses the U.S. commercial average price (proxy until the Virginia price is collected)

Case A debt sizing: computed once cost, OPEX and financing terms are sourced

Engine ready — rent KRW 250,000/kW-month, pre-lease 16–23.5% (base 19.8%), KTB 10y base

CFADS, debt service and DSCR *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Debt: sculpting vs gearing cap *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Repayment waterfall *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Results by demand scenario *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Stress: rent, occupancy, pre-lease, CAPEX, rates (±100bp), delay — breakevens computed once inputs are sourced

Case A in full, Case B summarised · breakevens at min DSCR 1.0x and target · tornado · heatmap (rent × CAPEX)

Tornado: min DSCR *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Breakevens (DSCR 1.0x and target) *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

DSCR heatmap: rent × CAPEX *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research

Case B stress summary *1

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Rent (per kW-month), Rent escalation, Pre-lease ratio, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation and 9 more

Source: *1 KSJ Research

Phase 2 (20 MW) is an option — values and trigger rent computed once inputs are sourced (ref.: phase-1 approval 3.0%)

Uncertainty: grid approval (probability), rent (stochastic), demand (IEA scenarios) — LSM Monte Carlo, fixed seed

Option value by approval probability *1,*2

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified)

Build now vs wait *1,*2

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified)

Monte Carlo convergence *1,*2

Generated once inputs are sourced

Inputs needed: CAPEX per MW, Construction period, Stabilised occupancy, OPEX (per kW-year, ex power), OPEX escalation, Combined tax rate, Depreciation life, Operating period and 7 more

Source: *1 KSJ Research *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified)

Decision structure: grid approval (p) → observe rent and demand → build now or wait *1

Phase 2 (20 MW) = the right to wait for grid approval and demand

- Uncertainty ①: grid approval with probability p at the decision date; zero value if refused
- Uncertainty ②: rent follows geometric Brownian motion (drift = escalation, volatility σ)
- Uncertainty ③: demand path drawn from IEA low/base/high growth — observed via phase 1 occupancy
- Phase 2 pre-lease = phase 1 occupancy in the decision year (build after demand is confirmed)
- Valuation: Monte Carlo + LSM (least-squares continuation value), fixed seed, convergence by path count

Source: *1 KSJ Research
Note: Unlevered NPV at the equity discount rate. Option value = p × value given approval

Korea vs U.S.: different bottlenecks set debt capacity — grid and pre-leasing in Korea, scale and rates in the U.S.

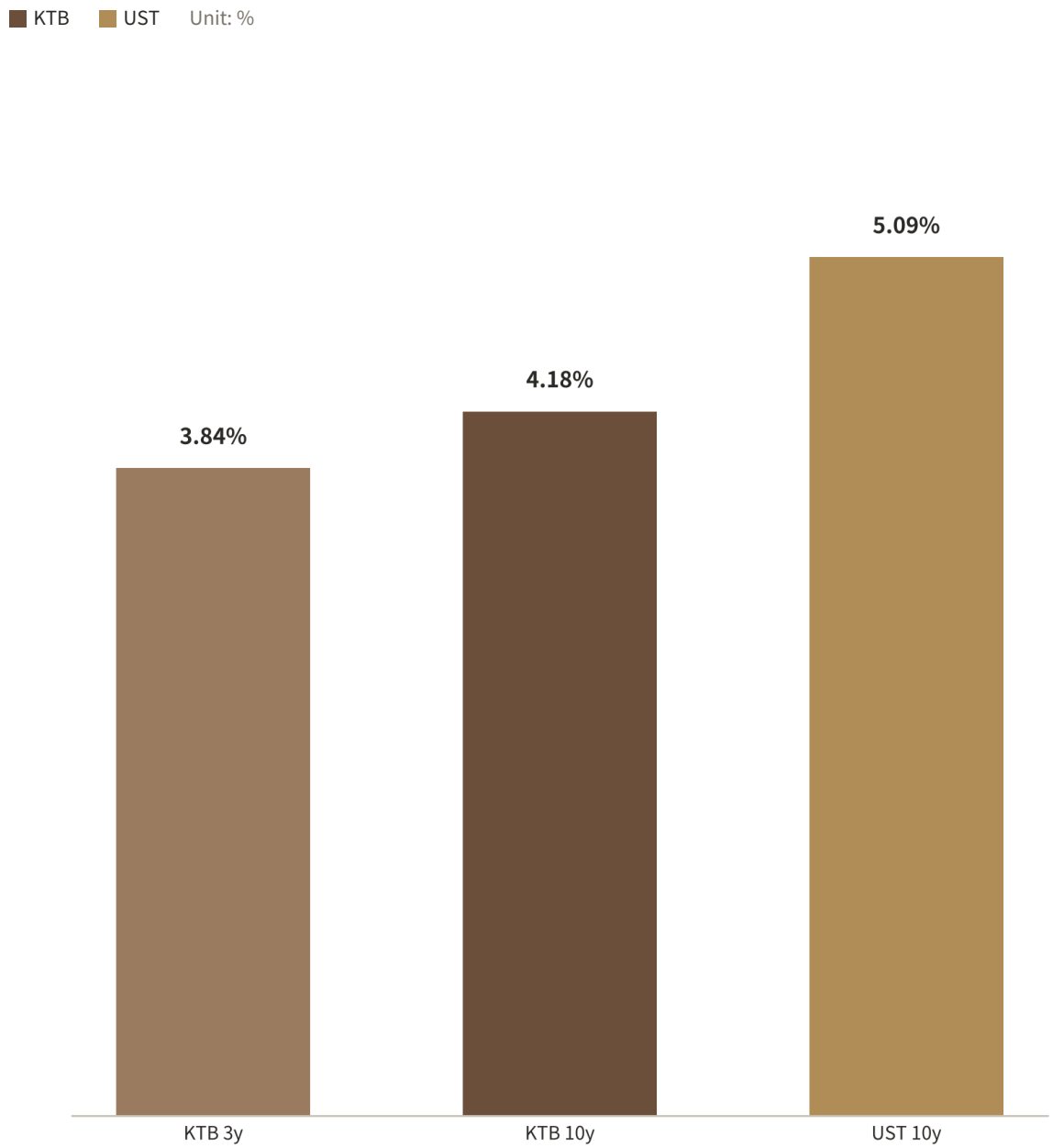
Base-rate gap +0.91pp (UST 10y − KTB 10y) · grid approval 3.0% · pre-leasing 16–23.5% · FX 1,332.1

Debt, DSCR and equity IRR side by side — same engine, different bottlenecks *1,*2,*3,*4,*5

	Korea 40 MW (Case A)	U.S. 100 MW (Case B)
Capacity · phasing	40 MW (20 firm + 20 option)	100 MW single
Currency · base rate	KRW · 4.18%	USD · 5.09%
Margin	being sourced	being sourced
Debt (local currency)	after inputs	after inputs
Debt (KRW)	after inputs	after inputs
Debt per MW (KRW)	after inputs	after inputs
Gearing · binding	after inputs	after inputs
Min · avg DSCR	after inputs	after inputs
LLCR	after inputs	after inputs
Project IRR · Equity IRR	after inputs	after inputs
Key bottleneck	**Grid (3.0% approved) · pre-leasing (16–23.5%)**	**Scale (100 MW, 5x Korea phase 1) · rates (+0.91pp)**

Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10) *3 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified) *4 Press citing market reports (rent, pre-leasing; source being verified) *5 KSJ Research
Note: KRW = USD mn × ECOS USD/KRW 1,332.1 (2026-10-20) ÷ 100. Korea debt is for phase 1 (20 MW)

Base rates: UST 10y 5.09% vs KTB 10y 4.18% — at the same DSCR, U.S. debt service is heavier *1,*2



Source: *1 Bank of Korea ECOS market rates (daily), USD/KRW *2 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: ECOS 2026-10-20, FRED 2026-10-16. Debt comparison after inputs

Implication: for Korean AI data-centre PF, 'when' comes before 'how much' — grid approval and an anchor tenant first

Lender checklist · limitations · assumptions awaiting sources · next: Vol.3 (IPO pricing, overseas peers)

Lender checklist: Korea — approval and tenants first; U.S. — rates and cost first [*1,*2,*3,*4,*5]

Item	Korea 40 MW	U.S. 100 MW	Pages
Grid approval	Phase 1 approval a must; phase 2 as an option (3.0%)	Confirm connection timing	1, 5
Pre-leasing	Market 16–23.5%; breakeven after inputs	Anchor tenant	3, 4
Rates	KTB 10y 4.18%	UST 10y 5.09% — hedging, margin	2, 6
Construction cost	Fixed-price EPC; equipment price clauses	Large absolute exposure	2, 4
Power cost	Pass-through clause; carve-outs	Tenant burden as prices rise	1, 2
Liquidity	DSRA; grace over ramp-up	DSRA	3

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified) *2 Press citing market reports (rent, pre-leasing; source being verified) *3 Bank of Korea ECOS market rates (daily), USD/KRW *4 FRED, Federal Reserve Bank of St. Louis (DGS10) *5 KSJ Research

Note: Checklist for the hypothetical cases; figures from the body

Limitations of this note

- Both cases are hypothetical; assumptions without a verified public source are not used
- Rent and pre-leasing figures come from press citing market reports, pending source checks
- Ramp-up is proxied by IEA global data-centre power growth; local absorption may differ
- Peer margin → price is an upper bound, not causal (volume, FX, mix not separated)
- FX hedging, tax details (loss-carryforward limits) and maintenance CAPEX are simplified
- Case B power price uses the U.S. commercial average until Virginia's is collected

Note: Updated as sources are confirmed

Next in Vol.3: IPO pricing and overseas peers

A hypothetical unlisted power-equipment IPO in a re-rated sector: what price range?

- Korean peers' P/E and EV/EBITDA + overseas peers (SEC EDGAR)
- Check that market cap (listed parent) and financials (filer) are the same entity
- Fair value → IPO discount → price range · Monte Carlo sensitivity

Note: Timing and method of Vol.3 may change

Assumptions awaiting sources: Case A 25, Case B 21

- No invented numbers: unsourced assumptions are left blank, not computed
- Once sourced, every table, chart and conclusion is recalculated by the same engine
- Assumptions, units, sources and status by case: Appendix A2–A3

Note: Status: input needed · verify · sourced · data · design · neutral

Expected Q&A: what lenders ask first

Questions likely in credit review and lender due diligence, with page references

8 expected questions, including the lender's view [*1,*2,*3,*4,*5,*6,*7]

Question	Answer	Page
Can it borrow without an anchor tenant?	Hard. Pre-leasing for new Seoul-metro supply is 16–23.5%, and filling at market growth (IEA base 10.1%/yr) takes 16.8 yrs. Lenders first ask whether contracted rent covers debt service; the breakeven pre-lease is computed on p4 once inputs are sourced	2, 3, 4
Why 40 MW in Korea?	1,010 MW approved of 33,592 MW applied in the Seoul metro, phase 1 (3.0%). Projects of 10 MW or more need a grid impact assessment, and domestic operating sites are around 40 MW. Hence 20 MW firm plus a 20 MW option built after approval and demand are confirmed	1, 5
Why 100 MW in the U.S., and what differs?	Virginia commercial sales grow 5.6%/yr vs U.S. 0.9%. But base rates are 0.91pp above KTB, so debt service is heavier at the same DSCR	2, 6
What if rates rise 100bp?	±100bp on base rates (KTB 10y 4.18%, UST 10y 5.09%); the DSCR impact is computed on p4 once margins are sourced	4
What if equipment prices rise?	If the three peers' margin rise 5.2%→14.4% (FY2022A→FY2025A) were all price, equipment prices rose 3.3%–23.9% (upper bound, not causal).	2, 4
Who pays for power?	A 100% pass-through: the tenant pays usage × tariff, so the SPC's net power cost is zero and tariff risk becomes a tenant-affordability question (U.S. commercial average 13.41¢/kWh, 2025)	1, 2
Build phase 2 now or wait?	Grid approval (ref.: phase-1 approval 3.0%) plus rent and demand uncertainty valued by LSM Monte Carlo; computed on p5 once inputs are sourced	5
How is FX handled for the U.S. case?	Computed in USD, converted at ECOS USD/KRW 1,332.1 (2026-10-20): KRW 100mn = USD mn × FX ÷ 100. Hedging costs are excluded	6, A4

Source: *1 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified) *2 Press citing market reports (rent, pre-leasing; source being verified) *3 IEA, Energy and AI (Apr 2025), executive summary *4 Bank of Korea ECOS market rates (daily), USD/KRW *5 FRED, Federal Reserve Bank of St. Louis (DGS10) *6 FSS DART annual reports (consolidated) *7 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Numbers match the body; shaded = core lender questions

Assumptions and sources: Korea 40 MW

Case A — 40 MW leased data centre SPC, outer Seoul metro (Incheon / western Gyeonggi)

Korea 40 MW: 25 of 34 assumptions await sources — unsourced numbers are not used [*1,*2]

Item	Value	Unit	Basis / source	Status
Phase 1 capacity	20	MW	Hypothetical case design	Design
Phase 2 (option) capacity	20	MW	Hypothetical case design	Design
CAPEX per MW	—	KRW 100mn/MW	Being sourced	Input needed
Power equipment share of CAPEX	—	%	Being sourced	Input needed
Construction period	—	quarters	Being sourced	Input needed
Rent (per kW-month)	250,000	KRW/kW-month	Press citing CBRE (2025)	Verify
Rent escalation	2.5% (2–3%)	%/yr	2–3% p.a. (mid 2.5%)	Verify
Pre-lease ratio	19.75% (16–23.5%)	%	Seoul-metro 2027–28 deliveries 16–23.5% (base = midpoint)	Verify
Stabilised occupancy	—	%	Being sourced	Input needed
Ramp-up elasticity	1	x	Neutral value pending evidence	Neutral
OPEX (per kW-year, ex power)	—	KRW/kW-yr	Being sourced	Input needed
OPEX escalation	—	%/yr	Being sourced	Input needed
PUE	—	x	Being sourced	Input needed
IT load factor	—	%	Being sourced	Input needed
Power price	—	KRW/kWh	Being sourced	Input needed
Power pass-through	100%	%	Hypothetical case design	Design
Corporate tax (national/federal)	—	%	Being sourced	Input needed
Local/state tax	10%	% of national tax	Va. Code § 58.1-400 (B) / local income tax (A)	Verify
Combined tax rate	—	%	Being sourced	Input needed
Depreciation life	—	yrs	Being sourced	Input needed
Operating period	—	yrs	Being sourced	Input needed
Exit cap rate	—	%	Being sourced	Input needed
Credit margin	—	bp	Being sourced	Input needed
Target DSCR	—	x	Being sourced	Input needed
Maximum gearing	—	%	Being sourced	Input needed
Loan tenor	—	yrs	Being sourced	Input needed
Grace period	—	yrs	Being sourced	Input needed
Grace-period interest	paid	-	Hypothetical case design	Design
DSRA months	—	months	Being sourced	Input needed
Discount rate (equity)	—	%	Being sourced	Input needed
Grid-approval probability	—	%	Being sourced	Input needed
Approval decision year	—	yrs after phase-1 COD	Being sourced	Input needed
Option window	—	yrs	Being sourced	Input needed
Rent volatility	—	%/yr	Being sourced	Input needed

Source: *1 Press citing market reports (rent, pre-leasing; source being verified) *2 Public reports on Seoul-metro grid applications (Aug 2024–Mar 2026; source being verified)
Note: Status: input needed = not yet sourced (excluded) · verify = public basis, not yet checked against the original · design = hypothetical case setting · neutral = neutral pending evidence · data = from official statistics

Assumptions and sources: U.S. 100 MW

Case B — 100 MW hyperscale leased data centre, Virginia, U.S.

U.S. 100 MW: 21 of 30 assumptions await sources — unsourced numbers are not used [*1,*2]

Item	Value	Unit	Basis / source	Status
Phase 1 capacity	100	MW	Hypothetical case design	Design
Phase 2 (option) capacity	0	MW	Hypothetical case design	Design
CAPEX per MW	—	USD mn/MW	Being sourced	Input needed
Power equipment share of CAPEX	—	%	Being sourced	Input needed
Construction period	—	quarters	Being sourced	Input needed
Rent (per kW-month)	—	USD/kW-month	Being sourced	Input needed
Rent escalation	—	%/yr	Being sourced	Input needed
Pre-lease ratio	—	%	Being sourced	Input needed
Stabilised occupancy	—	%	Being sourced	Input needed
Ramp-up elasticity	1	x	Neutral value pending evidence	Neutral
OPEX (per kW-year, ex power)	—	USD/kW-yr	Being sourced	Input needed
OPEX escalation	—	%/yr	Being sourced	Input needed
PUE	—	x	Being sourced	Input needed
IT load factor	—	%	Being sourced	Input needed
Power price	0.1341	USD/kWh	EIA commercial average price, U.S. (proxy), 2025	Data
Power pass-through	100%	%	Hypothetical case design	Design
Corporate tax (national/federal)	21%	%	26 U.S.C. § 11(b)	Verify
Local/state tax	6%	%	Va. Code § 58.1-400 (B) / local income tax (A)	Verify
Combined tax rate	25.74%	%	Public source, being verified	Verify
Depreciation life	—	yrs	Being sourced	Input needed
Operating period	—	yrs	Being sourced	Input needed
Exit cap rate	—	%	Being sourced	Input needed
Credit margin	—	bp	Being sourced	Input needed
Target DSCR	—	x	Being sourced	Input needed
Maximum gearing	—	%	Being sourced	Input needed
Loan tenor	—	yrs	Being sourced	Input needed
Grace period	—	yrs	Being sourced	Input needed
Grace-period interest	paid	-	Hypothetical case design	Design
DSRA months	—	months	Being sourced	Input needed
Discount rate (equity)	—	%	Being sourced	Input needed

Source: *1 Press citing market reports (rent, pre-leasing; source being verified) *2 26 U.S.C. § 11(b); Va. Code § 58.1-400
Note: Status: input needed = not yet sourced (excluded) · verify = public basis, not yet checked against the original · design = hypothetical case setting · neutral = neutral pending evidence · data = from official statistics

Data notes: sources, datasets, dates and model inputs

The same official-source data as Vol.1, linked to PF inputs · overseas peers (SEC) in Vol.3

Data sources · datasets · as-of dates

Source	Dataset	Items	As of	Status
IEA	Energy and AI (Apr 2025), Key Questions on Energy and AI (Apr 2026)	Data-centre power, 2024 and 2035 scenarios	2026-10-21	Collected
ECOS · Bank of Korea	Market rates (daily), KRW exchange rates	KTB 3y/10y, USD/KRW	2026-10-20	Collected
FRED · St. Louis Fed	10-Year Treasury Constant Maturity (DGS10)	UST 10y	2026-10-20	Collected
DART · FSS	Annual reports, consolidated	Three peers' revenue and operating income	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (state/sector, annual)	Virginia and U.S. commercial sales and price	2026-10-20	Collected
EIA · U.S. EIA	Electricity retail sales (Virginia commercial price)	Virginia commercial average price	—	Not collected

Note: As-of = collection date (IEA: date checked against source). Overseas peers (SEC) are used in Vol.3

Data values used in the model, with dates

Item	Value	Date / period	Source
KTB 3y · 10y	3.840% · 4.176%	2026-10-20	ECOS
UST 10y (DGS10)	5.09%	2026-10-16	FRED
USD/KRW	1,332.1	2026-10-20	ECOS
Demand growth (low·base·high)	4.87% · 10.13% · 13.68%	2024→2035	IEA
3-peer combined OPM	5.16% → 14.44%	FY2022A→FY2025A	DART
Grid approval ratio	1,010 ÷ 33,592 = 3.01%	2024.8~2026.3	Public reports (verifying)
Commercial average price	13.41¢/kWh (US)	2025	EIA

Note: Latest business day on or before 2026-10-20. KRW = USD mn × USD/KRW ÷ 100

Methodology: one engine, case parameters, verified calculations

Every figure is calculated from data and assumptions, and conclusions are generated from the results

Data links → PF model → stress and option → checks and report [*1]

1. Data links

IEA → ramp-up

ECOS · FRED → base rates

DART → CAPEX stress

EIA → Case B

2. PF model

Quarterly build, IDC

CFADS → sculpting

Gearing cap

Waterfall · DSRA

3. Stress, option

Breakevens, tornado

Heatmap

LSM Monte Carlo

Fixed seed, convergence

4. Checks, report

Accounting identities

DSCR, IRR, FX checks

Conclusion titles

Korean & English

Built with an AI coding tool (Claude Code) for data links, the PF model, Monte Carlo and report generation; case design, assumption sourcing and number checks by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Glossary: the lender's vocabulary

Units: KRW 100mn, USD mn · %, x

Glossary

Term	Definition
CFADS	Cash flow available for debt service = EBITDA − tax
DSCR	Debt service cover ratio = CFADS ÷ (interest + principal)
LLCR	Loan life cover ratio = PV of CFADS over loan life ÷ debt
Sculpting	Sizing debt by setting each year's debt service = CFADS ÷ target DSCR
Maximum gearing	Cap on debt ÷ total project cost (LTC)
DSRA	Debt service reserve account: months of next year's debt service
IDC	Interest during construction, capitalised into project cost
Grace period	Principal holiday (interest paid or capitalised)
Committed lease	Long-term lease paying for contracted capacity regardless of use
Pass-through	Clause passing power cost to the tenant at usage × tariff
MW (IT)	Capacity for IT equipment; total facility = IT × PUE
PUE	Total facility power ÷ IT power
Project IRR / Equity IRR	IRR of unlevered after-tax cash flow / of equity cash flow
Real option	Value of the right to defer an investment
LSM	Least-squares Monte Carlo: regression estimates continuation value to time exercise
Grid impact assessment	Pre-assessment of grid impact for large loads (10 MW+)

Note: Shaded = core lender metrics

Disclosures

- This is a personal portfolio for education and research. It is not investment or financial advice.
- Both cases are hypothetical, not real projects; the same model engine runs with different assumptions.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Assumptions and their sources are in Appendix A2–A3.
- As of 2026-10-20. Methods and code can be reproduced from the repository.

KSJ Research

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio